RESEARCH POSTER / FOUNDATIONAL MACHINE LEARNING / 2017

# ATTENTION IS ALL YOU NEED

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THE CENTRAL IDEA

## NO RECURRENCE. ALL ATTENTION.

Model global token interactions directly, while training positions in parallel.

## 01 / WHY REPLACE RECURRENCE?

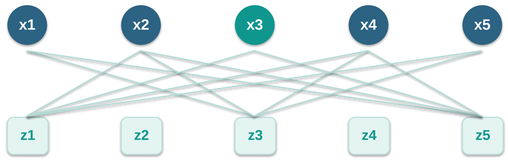
**Recurrent computation requires sequential state updates.**

RECURRENT / SEQUENTIAL




Dependency depth grows with sequence length n

SELF-ATTENTION / PARALLEL




Every position can attend to other positions in one layer.

## 02 / THE ATTENTION OPERATION

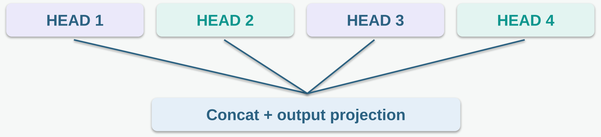
Scaled dot-product attention

$$\text{softmax}(QK^T / \sqrt{d_k})\, V$$

Q = queries · K = keys · V = values

## 03 / MULTI-HEAD ATTENTION

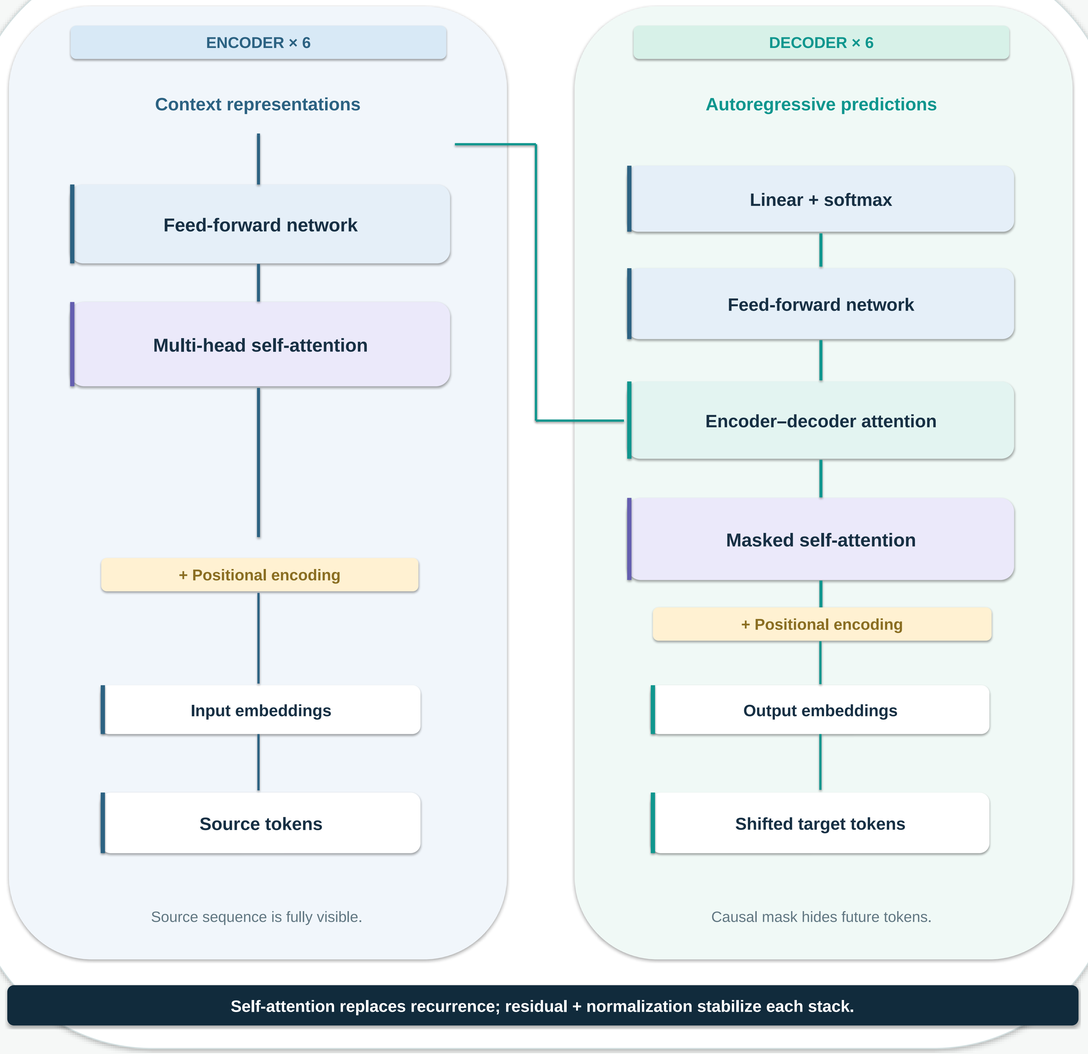
Independent learned projections capture varied relationships.




Four heads illustrated; the base model has eight.

## 04 / THE TRANSFORMER, IN ONE VIEW

Six encoder layers and six decoder layers.




Source sequence is fully visible.

Causal mask hides future tokens.

Self-attention replaces recurrence; residual + normalization stabilize each stack.

## 05 / POSITIONAL ENCODING

Sinusoidal position signals are added to token embeddings.



## 06 / EVIDENCE FROM TRANSLATION

WMT14 · EN → DE

**28.4**

BLEU / Transformer (big)

2017 NeurIPS proceedings · Table 2

WMT14 · EN → FR

**41.0**

BLEU / Transformer (big)

2017 NeurIPS proceedings · Table 2

## 07 / BASE-MODEL BLUEPRINT

| ENCODER / DECODER | ATTENTION HEADS |
|---|---|
| 6 / 6 | 8 |
| **MODEL WIDTH** | **FEED-FORWARD** |
| 512 | 2,048 |
| **PARAMETERS** | **KEY WIDTH** |
| ≈65M | 64 |

## 08 / WHAT ACTUALLY IMPROVES?

**GLOBAL CONTEXT. PARALLEL TRAINING.**

Short attention paths across sequence positions.

IMPORTANT LIMITATION

Dense self-attention has quadratic pairwise token interactions. The decoder still emits output tokens autoregressively.

SOURCE / Vaswani et al. (2017), Attention Is All You Need. NeurIPS 30, Table 2.
Educational architecture redraw · NeurIPS EN–FR 41.0 differs from later arXiv revisions · https://arxiv.org/abs/1706.03762